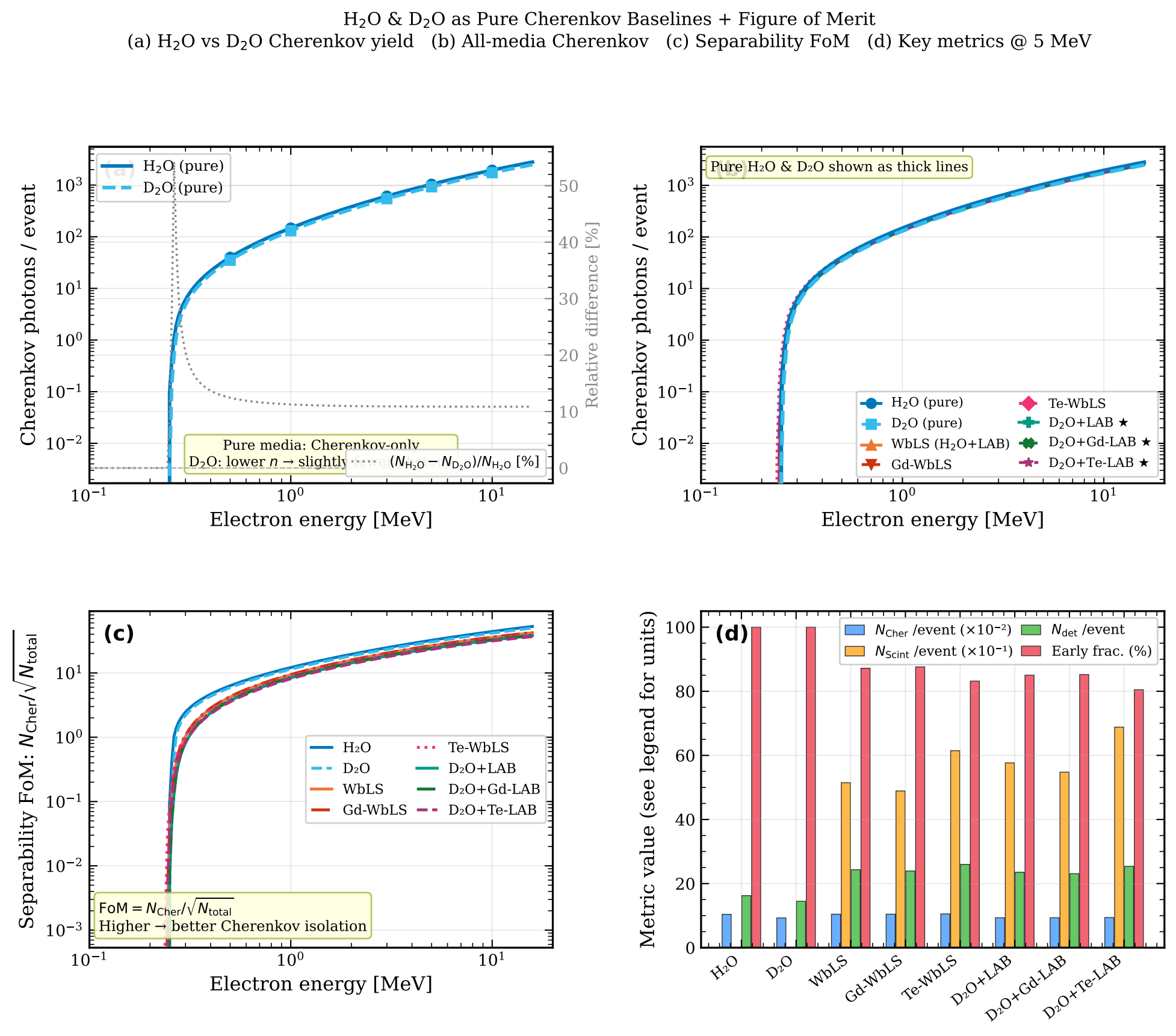

H₂O & D₂O as Pure Cherenkov Baselines + Figure of Merit
(a) H₂O vs D₂O Cherenkov yield (b) All-media Cherenkov (c) Separability FoM (d) Key metrics @ 5 MeV
(a)
H₂O (pure)
D₂O (pure)
Cherenkov photons / event
Relative difference [%]
Pure media: Cherenkov-only
D₂O: lower n → slightly
(N_H₂O − N_D₂O)/N_H₂O [%]
Electron energy [MeV]
(b)
Pure H₂O & D₂O shown as thick lines
H₂O (pure)
D₂O (pure)
WbLS (H₂O+LAB)
Gd-WbLS
Te-WbLS
D₂O+LAB ★
D₂O+Gd-LAB ★
D₂O+Te-LAB ★
Cherenkov photons / event
Electron energy [MeV]
(c)
Separability FoM: N_Cher/√N_total
H₂O
D₂O
WbLS
Gd-WbLS
Te-WbLS
D₂O+LAB
D₂O+Gd-LAB
D₂O+Te-LAB
FoM = N_Cher/√N_total
Higher → better Cherenkov isolation
Electron energy [MeV]
(d)
N_Cher /event (×10⁻²)
N_det /event
N_Scint /event (×10⁻¹)
Early frac. (%)
Metric value (see legend for units)
H₂O
D₂O
WbLS
Gd-WbLS
Te-WbLS
D₂O+LAB
D₂O+Gd-LAB
D₂O+Te-LAB